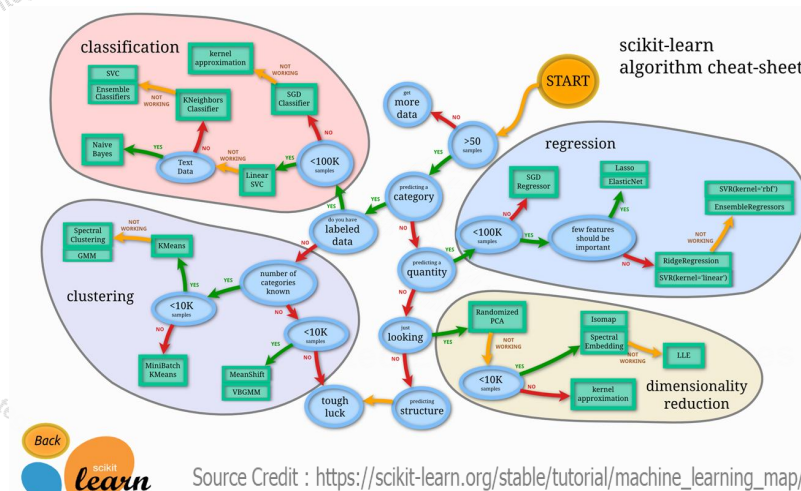


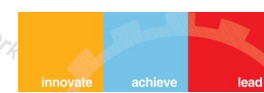
Model Guide



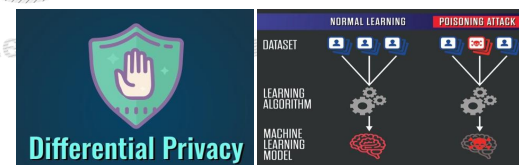
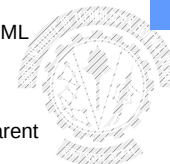
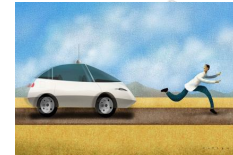
Source Credit : https://scikit-learn.org/stable/tutorial/machine_learning_map/index.html

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Emerging Requirements

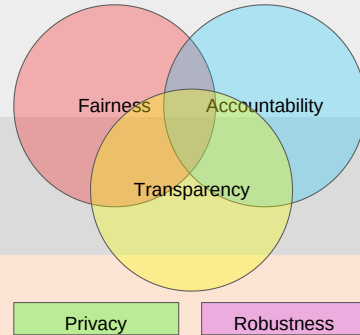


- **Fairness**
 - What to do to ensure gender and ethnic fairness in ML models?
- **Accountability**
 - Who takes the responsibilities for failed ML models?
- **Transparency**
 - What to do to make ML models transparent and comply with regulations?
- **Privacy Issues**
 - How to protect user privacies when exposing data to ML models?
- **Security Issues**
 - How do we defend ML models against data poisoning?



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FAccT Overview (More on this is a part of next semester elective)

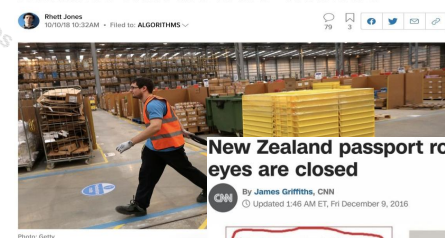


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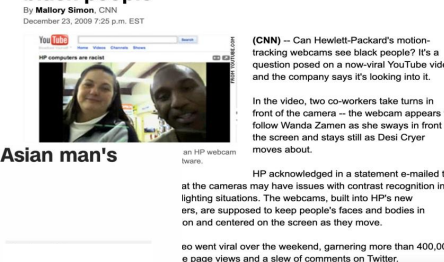
Real World Example



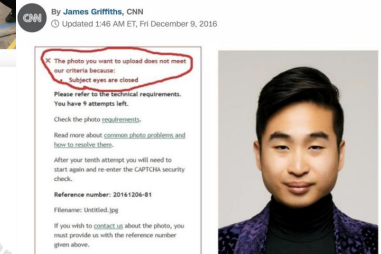
Amazon's Secret AI Hiring Tool Reportedly 'Penalized' Resumes With the Word 'Women's'



HP looking into claim webcams can't see black people



New Zealand passport robot thinks this Asian man's eyes are closed



New Zealand's online passport application system couldn't recognize Richard Lee's open eyes.

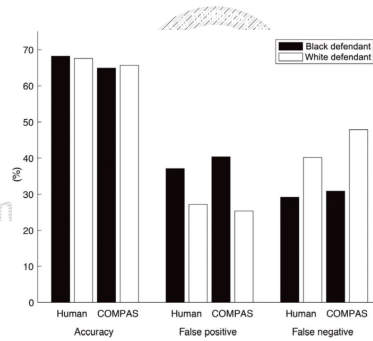
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Algorithmic Bias



Commercial risk assessment software known as COMPAS

- Assess more than 1 million offenders since 2000
- Predicts a defendant's risk of committing a misdemeanor or felony
- 137 features



[Dressel et al, 2018](#)

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Bias in Historical Data – Negative Legacy



[Gard et al, 2018](#)

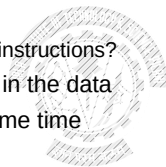
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Fairness



<https://www.borealisai.com/research-blogs/tutorial1-bias-and-fairness-ai/>

- What is Fairness?
 - The absence of bias towards an individual or a group ([Mehrabi et al, 2019](#))
- Can ML Models Discriminate?
 - Aren't machines just follow human's instructions?
 - ML models approximate patterns in the data
 - Learns/Amplifies biases at the same time



Work Integrated Learning Programmes

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Fairness Through Unawareness



- A ML Algorithm Achieves Fair Through Unawareness If
 - None of the sensitive features are directly used in the model

Race and Ethnicity	Skills	Years of Exp	Hired?
Hispanic	Javascript	1	no
Hispanic	C++	5	yes
White	Java	2	yes
White	C++	3	yes

Training



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Issues With Fairness Through Unawareness

Sensitive Features May Still Be Used

- Inferred from indirect evidence

• Protected

• Inferred

Race and Ethnicity	Skills	Years of Exp	Often Goes to Mexican Markets	Hiring Decision		
Hispanic	Javascript	1	yes	no	Training	Discriminatory ML Model
Hispanic	C++	5	yes	yes		
White	Java	2	no	yes		
White	C++	3	no	yes		

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Limitations

Processing Sensitive Features

- Fairness through unawareness requires sensitive features to be masked out
- Not easy to do in real life
- Referred to as individual fairness criteria



Stereotypical dataset

The physician hired the secretary because he was overwhelmed with clients.

The physician hired the secretary because she was highly recommended.

Anti-stereotypical dataset

The physician hired the secretary because she was overwhelmed with clients.

The physician hired the secretary because he was highly recommended.

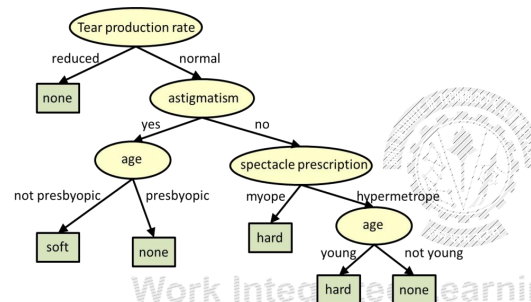
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Need for ML Interpretability

- Our society has been shifted to rely on AI more than ever
 - autonomous vehicles
 - Security
 - Finance
 - many others
- Who will benefit from ML Interpretability?
 - End Users: enhance trust, understand the consequences of the decisions, e.g., privacy, fairness.
 - Regulatory Agencies: compliance, audits, and accountability.
 - Model Designers: diagnose model performance

Interpretability in Decision Tree

- ML interpretability allows one to examine model's basis in its decision making process

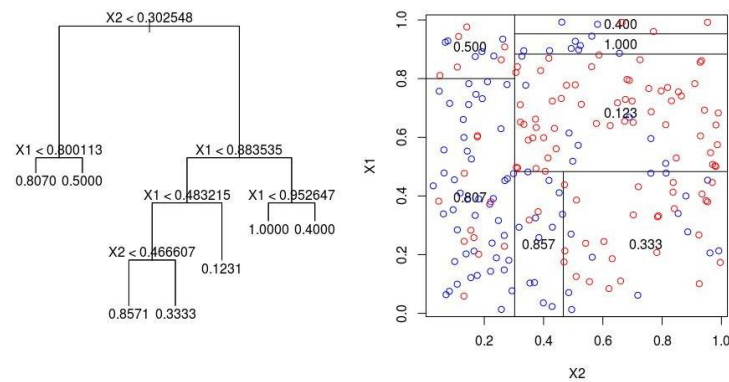


An interpretable tree model to find out the kind of contact lens a person may wear

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Interpretability in Decision Trees



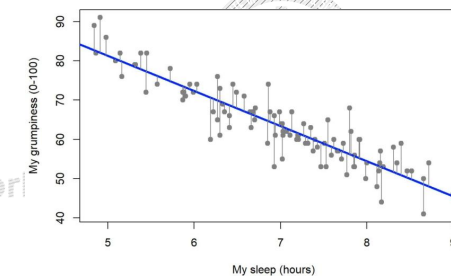
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Interpretability in Linear Regression



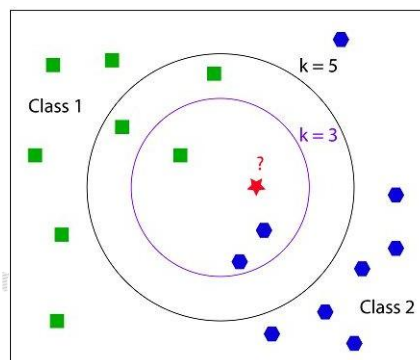
$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n,$$

interpretable components



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Interpretability in K-Nearest Neighbors



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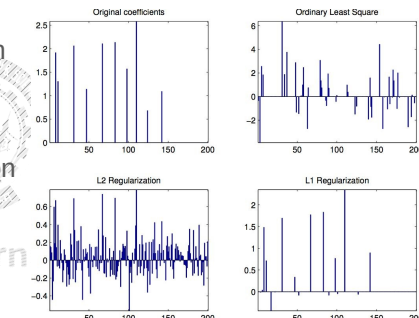
Sparsity for Interpretable Linear Regression



- In the case of linear regression
 - $\hat{y} = w_1 x_1 + w_2 x_2 + \dots + w_N x_N + b$
- Linear regression with L1 regularization
- Linear Regression with L2 regularization

$$Loss = Error(y, \hat{y}) + \lambda \sum_{i=1}^N |w_i|$$

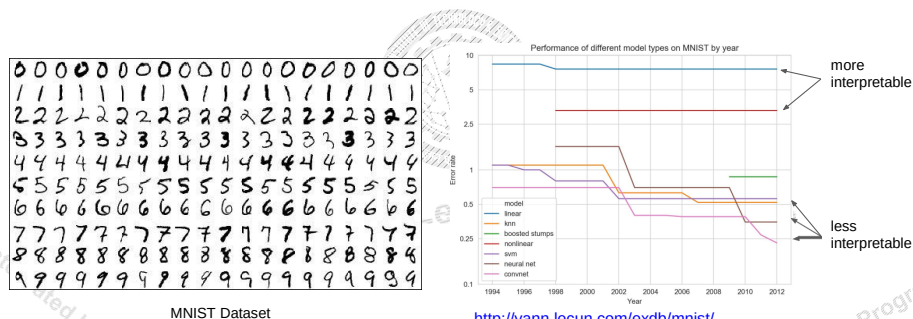
$$Loss = Error(y, \hat{y}) + \lambda \sum_{i=1}^N w_i^2$$



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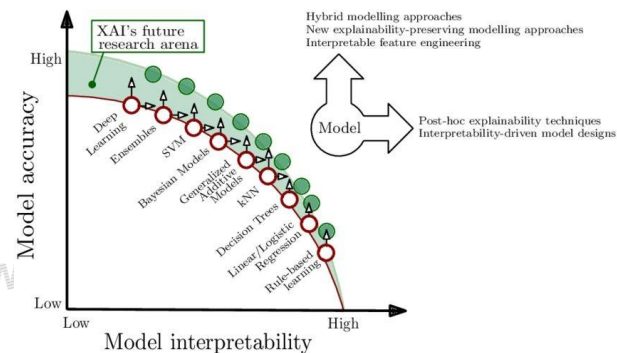
Interpretability and Performance Trade-offs

- Highly performed models tend to be less interpretable.
- Can powerful models with complex structures be interpretable at the same time?



<http://yann.lecun.com/exdb/mnist/>
<https://soph.info/2018/11/08/mnist-history/>

Interpretability and Performance Trade-offs



[Arrieta et al., 2019](#)